

BCS 404 Project Report

ACCRA TECHNICAL UNIVERSITY

Department of Computer Science

BCS 404: Introduction to Data Science with Python

Exploratory Data Analysis, Statistical Analysis and Machine Learning using a Real-World Dataset

A Project Report on the Titanic Dataset

Student Name: *[Your Name]*

Index Number: *[Your Index Number]*

Lecturer: Dr. Joseph Dadzie

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1. Introduction

Data science combines programming, statistics and domain knowledge to extract insight from data. This project demonstrates practical competence in the data-science workflow using Python and its scientific ecosystem — Pandas, NumPy, Matplotlib, Seaborn and Scikit-Learn — inside a Jupyter Notebook.

The dataset chosen is the famous **Titanic dataset** from Kaggle, which records the passengers of the RMS Titanic, the British ocean liner that sank in the North Atlantic on 15 April 1912 after striking an iceberg. More than 1,500 of the roughly 2,224 people on board died, making it one of the deadliest peacetime maritime disasters in history. The dataset is ideal for an introductory project because it combines numerical and categorical variables, contains realistic data-quality problems (missing values), and offers a natural binary classification task: **predicting whether a passenger survived**.

The objectives of the project are to:

1. Acquire and import the dataset into Python (Task 1);
2. Clean the data — detect and handle missing values and duplicates (Task 2);
3. Produce and interpret six visualisations (Task 3);
4. Perform descriptive, frequency and correlation analysis (Task 4);
5. Build and evaluate a Logistic Regression model that predicts survival (Task 5);
6. Discuss the findings, limitations and recommendations (Task 6).

2. Dataset Description

The dataset (`train.csv`) was downloaded from the Kaggle Titanic competition (<https://www.kaggle.com/competitions/titanic/data>). It contains **891 observations (passengers)** and **12 variables**:

Variable	Type	Description
PassengerId	Integer	Unique identifier for each passenger
Survived	Integer (0/1)	Target variable: 1 = survived, 0 = did not survive
Pclass	Integer (1/2/3)	Ticket class (1st, 2nd, 3rd) — a proxy for socio-economic status
Name	Text	Passenger name

Variable	Type	Description
Sex	Categorical	male / female
Age	Float	Age in years (fractional for infants)
SibSp	Integer	Number of siblings/spouses aboard
Parch	Integer	Number of parents/children aboard
Ticket	Text	Ticket number
Fare	Float	Passenger fare (£)
Cabin	Text	Cabin number
Embarked	Categorical	Port of embarkation: S = Southampton, C = Cherbourg, Q = Queenstown

3. Methodology

The project was carried out in a Jupyter Notebook using Python 3 with the following steps:

Data acquisition. The CSV file was loaded with `pandas.read_csv()`. The dimensions (`.shape`), column names (`.columns`), first five rows (`.head()`) and data types (`.dtypes`) were inspected.

Data cleaning. Missing values were detected with `df.isnull().sum()`. The treatment was: Age (177 missing, 19.9%) imputed with the **median** (28 years) because the age distribution is right-skewed and the median is robust to outliers; Cabin (687 missing, 77.1%) **dropped** because a column that is over three-quarters empty cannot be imputed reliably; Embarked (2 missing, 0.2%) imputed with the **mode** ('S'). Duplicates were checked with `df.duplicated()`; none were found.

Visualisation. Six figures were produced with Matplotlib and Seaborn, each with a title, axis labels and a written interpretation: an age histogram, a class-distribution bar chart, a boxplot of age by class, a scatter plot of age versus fare, a correlation heatmap, and a pairplot of selected numerical variables.

Statistical analysis. Descriptive statistics (`.describe()`), frequency distributions (`.value_counts()`), and a Pearson correlation matrix (`.corr()`) were computed. The strongest positive and negative correlations were identified programmatically from the upper triangle of the correlation matrix.

Machine learning. Seven predictors were selected (Pclass, Sex, Age, SibSp, Parch, Fare, Embarked). Sex and Embarked were label-encoded. The data were split 80% training / 20% testing with stratification (`train_test_split, random_state=42`). A **Logistic Regression** classifier (Scikit-Learn, `max_iter=1000`) was trained and evaluated on the test set using accuracy, the confusion matrix and the classification report.

4. Results

4.1 Data Acquisition

The dataset loaded successfully: **891 rows × 12 columns**, mixing numeric variables (Age, Fare, SibSp, Parch, Pclass, Survived, PassengerId) with categorical/text variables (Name, Sex, Ticket, Cabin, Embarked).

4.2 Data Cleaning

Column	Missing	%	Treatment
Age	177	19.87%	Imputed with median (28.0)
Cabin	687	77.10%	Column dropped
Embarked	2	0.22%	Imputed with mode ('S')

No duplicated rows were found (`df.duplicated().sum() = 0`). The cleaned dataset has 891 rows × 11 columns with no missing values.

4.3 Data Visualisation

Figure 1 — Histogram of passenger ages. Most passengers were young adults aged 20–40; the distribution is right-skewed with few passengers above 60. The spike at 28 partly reflects the median imputation.

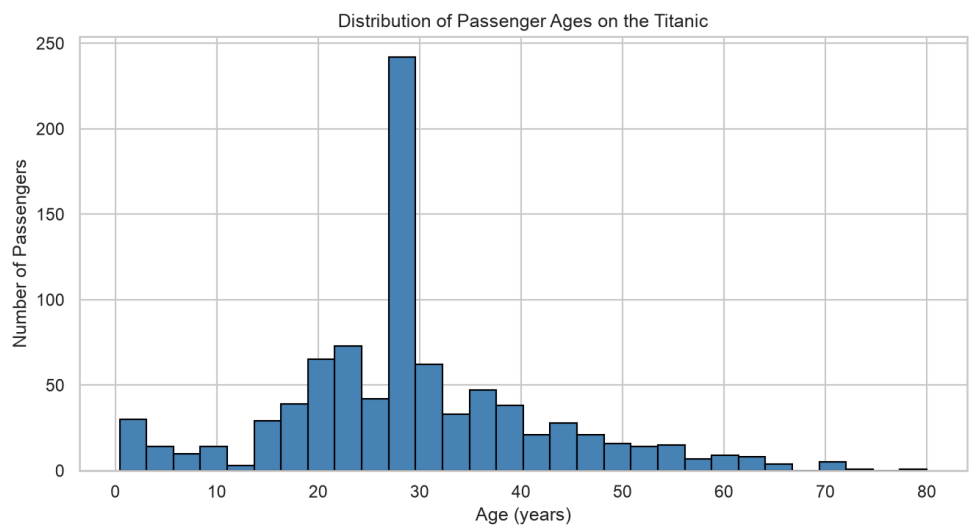


Figure 1: Distribution of Passenger Ages

Figure 2 — Passenger class distribution. Third class was the largest group (491 passengers, 55.1%), followed by first (216) and second (184).

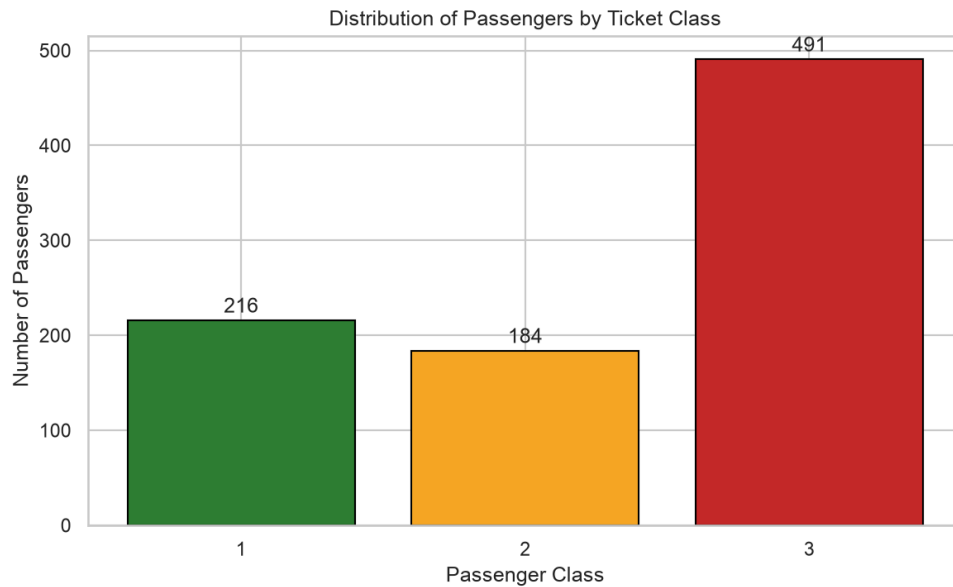


Figure 2: Distribution of Passengers by Ticket Class

Figure 3 — Age by passenger class. First-class passengers were oldest (median ≈ 35 years); second and third class had medians around 28. Wealth accumulates with age.

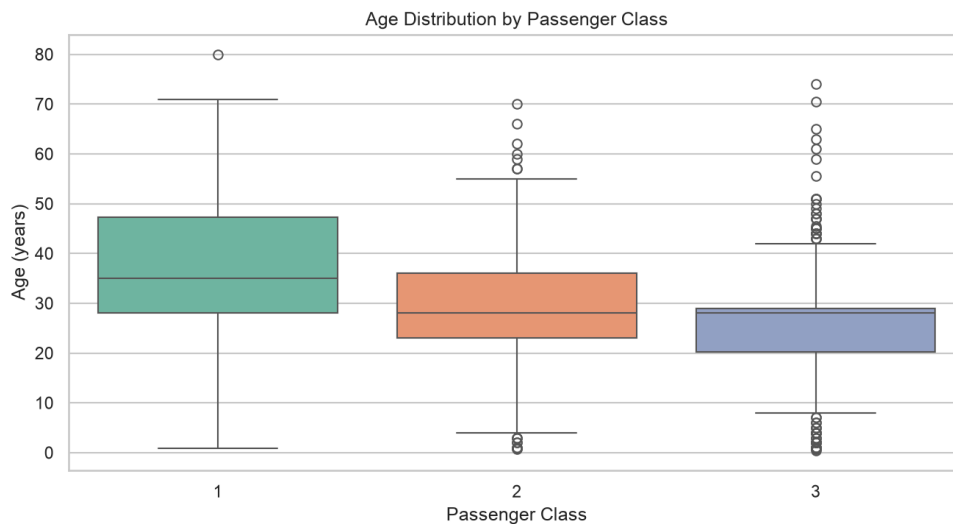


Figure 3: Age Distribution by Passenger Class

Figure 4 — Age versus fare. No strong linear relationship; a few first-class fares exceed £200 (maximum £512.33). Coloured by survival, high-fare passengers survived noticeably more often.

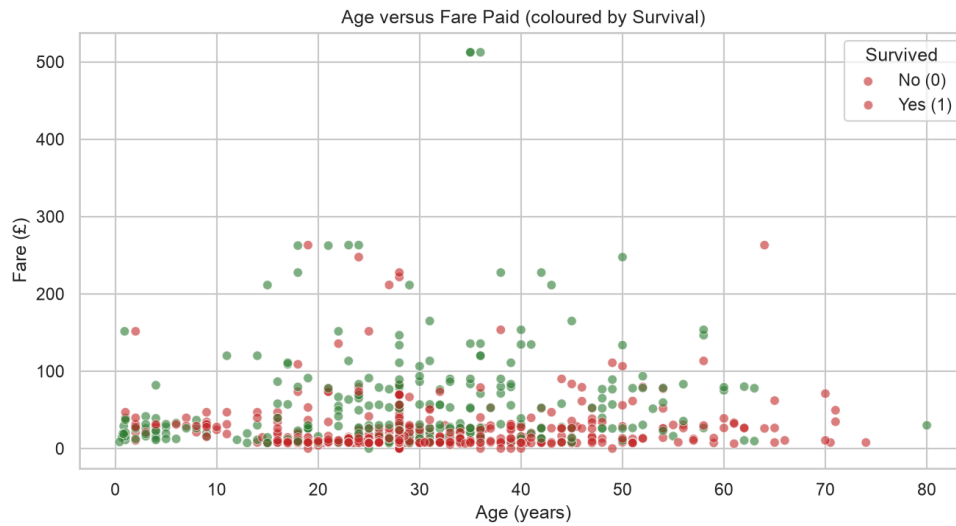


Figure 4: Age versus Fare Paid

Figure 5 — Correlation heatmap. Strongest relationships: Pclass–Fare ($r = -0.55$) and SibSp–Parch ($r = +0.42$). For survival: Pclass ($r = -0.34$) and Fare ($r = +0.26$).

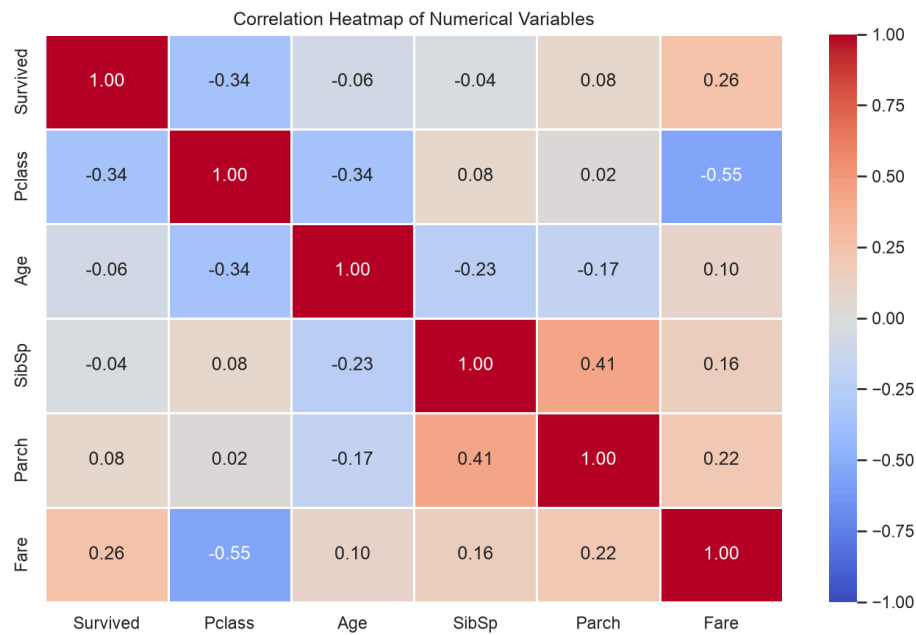


Figure 5: Correlation Heatmap of Numerical Variables

Figure 6 — Pairplot. Confirms the skewed distributions of Fare, SibSp and Parch; the clearest visual separation between survivors and non-survivors occurs along the fare axis.

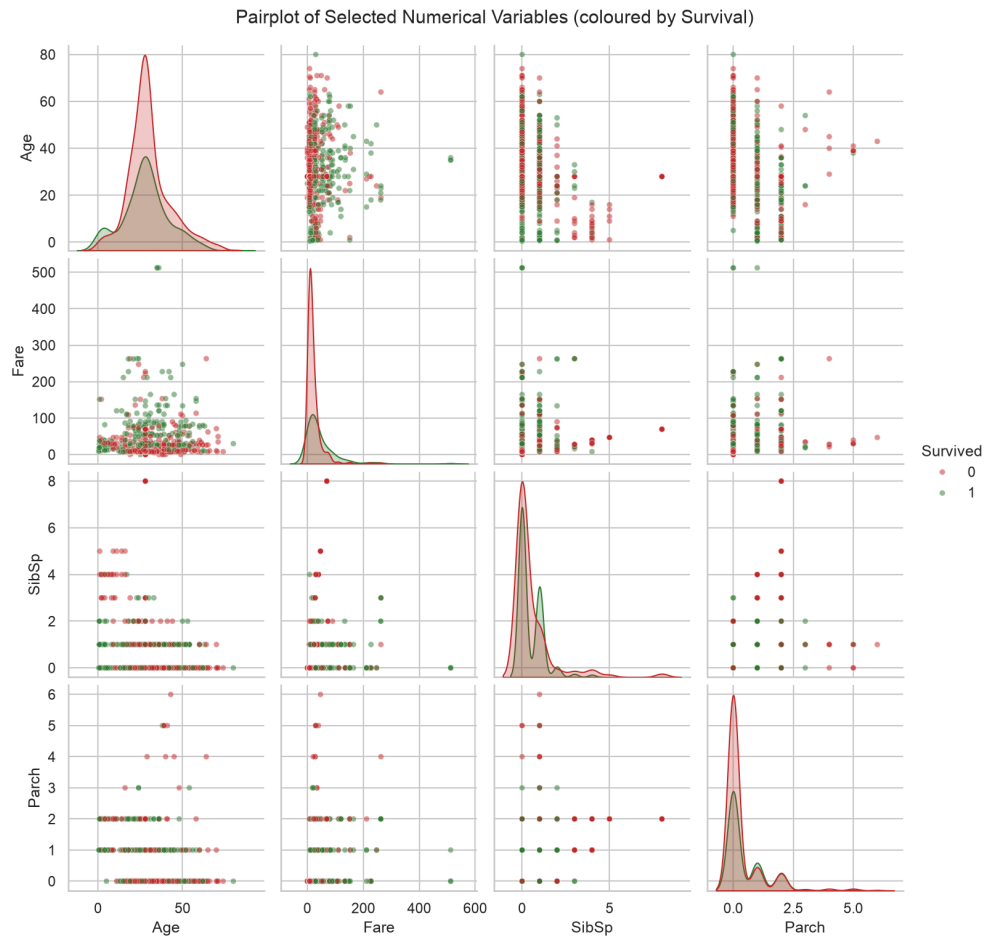


Figure 6: Pairplot of Selected Numerical Variables

4.4 Statistical Analysis

Descriptive statistics. Mean age 29.4 years (median 28); mean fare £32.20 but median only £14.45 (maximum £512.33) — a highly unequal fare distribution. Mean of Survived = 0.384, i.e. an overall survival rate of **38.4%**.

Frequency distributions. Died 549 (61.6%) vs survived 342 (38.4%); males 577 (64.8%) vs females 314 (35.2%); classes 3/1/2 = 491/216/184; embarked S/C/Q = 646/168/77 (72.5% / 18.9% / 8.6%).

Correlation analysis.

- **Strongest positive correlation: SibSp ↔ Parch, $r = +0.415$** — families travelled together.
- **Strongest negative correlation: Pclass ↔ Fare, $r = -0.549$** — higher classes (lower class numbers) paid higher fares.

Three important statistical findings:

1. **Sex was the strongest determinant of survival:** 74.2% of females survived versus only 18.9% of males — a four-fold difference reflecting the “women and children first” evacuation protocol.
2. **Socio-economic status strongly affected survival:** survival fell from 63.0% (1st class) to 47.3% (2nd) to 24.2% (3rd); consistent with Survived–Pclass $r = -0.34$ and Survived–Fare $r = +0.26$.

3. **Age mattered far less than expected:** Age–Survived correlation is only -0.065 ; for adults, sex and class — not age — decided survival.

4.5 Machine Learning Model

A Logistic Regression classifier was trained on 712 passengers and tested on 179.

- **Accuracy: 0.8045 (80.45%)** — well above the 61.6% majority-class baseline.

Confusion matrix (test set, n = 179):

	Predicted: Did not survive	Predicted: Survived
Actual: Did not survive	98 (TN)	12 (FP)
Actual: Survived	23 (FN)	46 (TP)

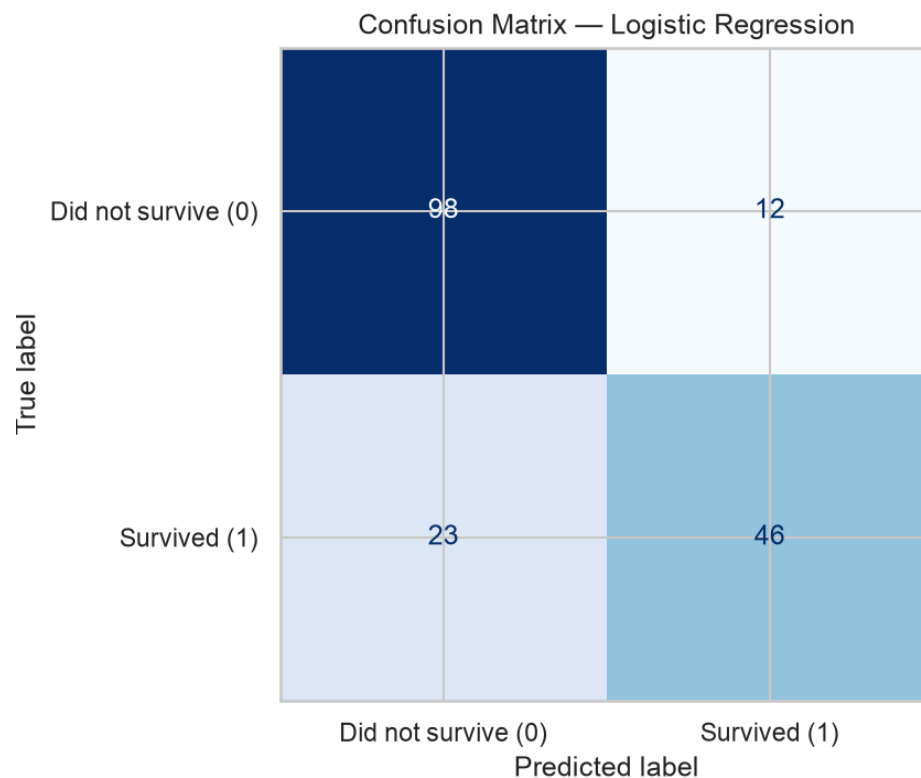


Figure 7: Confusion Matrix — Logistic Regression

Classification report:

Class	Precision	Recall	F1-score	Support
Did not survive (0)	0.81	0.89	0.85	110
Survived (1)	0.79	0.67	0.72	69
Accuracy			0.80	179
Weighted avg	0.80	0.80	0.80	179

The model detects non-survivors better (recall 0.89) than survivors (recall 0.67). The coefficients agree with the EDA: Sex (+2.56) and Pclass (−1.10) dominate the prediction.

5. Discussion

Major findings. Survival on the Titanic was not random. Although only 38.4% of the 891 passengers in the dataset survived, this overall rate conceals enormous group differences. Sex was the strongest factor — 74.2% of women survived against 18.9% of men — followed by passenger class, with survival falling steadily from 63.0% in first class to 24.2% in third. These patterns mirror the historical record: the “women and children first” order and the physical location of cabins, with third-class berths deep in the ship, far from the boat deck. The ship’s population was young (median 28 years), mostly male (64.8%), mostly travelling alone, predominantly third class (55.1%), and 72.5% boarded at Southampton.

Statistical insights. The correlation analysis quantified these relationships. The strongest positive correlation, SibSp–Parch ($r = +0.415$), shows that families boarded together. The strongest negative correlation, Pclass–Fare ($r = -0.549$), confirms that fare is essentially the monetary expression of class. Regarding survival, Pclass ($r = -0.34$) and Fare ($r = +0.26$) were the most informative numeric variables, while Age was almost uncorrelated with survival ($r = -0.065$) — a genuinely surprising result showing that age mattered only at the extremes (young children), whereas for adults wealth and sex decided everything. The fare statistics (mean £32.20 vs median £14.45, maximum £512.33) expose the sharp wealth divide aboard.

Machine learning results. The Logistic Regression model achieved 80.45% accuracy on unseen data — about four out of five passengers classified correctly, a substantial gain over the naive baseline. It was more reliable at identifying deaths (recall 0.89) than survivals (recall 0.67), partly due to class imbalance. Crucially, the model’s coefficients (female +2.56, class −1.10 per step) *validate* the statistical findings rather than contradict them, illustrating how EDA and machine learning reinforce one another.

Limitations. (1) The data cover 891 of ~1,309 known passengers and exclude the crew. (2) Median imputation of 177 ages concentrates the distribution artificially around 28 and may weaken age effects. (3) Dropping Cabin discarded deck-location information, and Name /Ticket were not mined for features. (4) Logistic Regression assumes linear log-odds and cannot capture interactions unless engineered. (5) Label-encoding Embarked imposes an artificial ordering. (6) No cross-validation or hyperparameter tuning was performed.

Recommendations. Future work should engineer features (titles from Name, FamilySize = SibSp + Parch + 1, deck letters from Cabin), one-hot encode Embarked, standardise numeric features, impute age by group medians, address class imbalance with class weights, and compare stronger models (Random Forest, Gradient Boosting) using k-fold cross-validation.

6. Conclusion

This project demonstrated the complete data-science workflow on a real-world dataset: acquisition, cleaning, visualisation, statistical analysis and predictive modelling. The evidence is unambiguous — **who a passenger was (female or male) and what they paid (first or third class) largely determined whether they survived the Titanic disaster**. A simple, interpretable Logistic Regression model captured these patterns well enough to predict survival with 80% accuracy, confirming that even introductory machine-learning tools can extract genuine insight from historical data.

7. References

1. Kaggle. (2012). *Titanic — Machine Learning from Disaster*. Retrieved from <https://www.kaggle.com/competitions/titanic/data>
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3. Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
4. Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95.
5. Waskom, M. (2021). seaborn: statistical data visualization. *Journal of Open Source Software*, 6(60), 3021.
6. Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with NumPy. *Nature*, 585, 357–362.

8. Appendix (Python Code)

The complete, executed Python code is contained in the accompanying Jupyter Notebook **Titanic_Data_Science_Project.ipynb** in the GitHub repository. For the printed report, export the notebook code (File → Save and Export Notebook As → PDF/HTML) and attach it here, or print the notebook itself as the appendix.